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The laboratory report

The purpose of a laboratory report is to communicate the aim, process and outcome of an investigation to an outside audience. It is a record of your direct (“hands-on”) experience in the laboratory. In most cases, a scientific investigation is considered to be incomplete without a report. By synthesising (putting together) the different aspects of your laboratory experience in a structured and coherent report, the essence of your investigation becomes clearer in your own mind. In the process, you develop your skills of reasoning and ability to communicate in writing.

There are many acceptable ways of presenting a scientific laboratory report, but, almost all reports will include the components outlined below.

F.1 Components of a laboratory report

Title:

This will include the author’s name (your name), your partners’ names, the course code, the date of the experiment and a suitable title of the report. The title must be short but factual and descriptive. It must summarise the major aspects to be dealt with in the report. The key words will often come from the laboratory task that has been set and you need to identify these. These words help clarify the requirements of the task and also alert the reader as to what the report is about.

Introduction and aim:

The introduction puts the report into perspective by giving the reader relevant background information about the phenomenon being investigated. This is in order to prepare the reader for what he or she is about to read. This background may include some historical information or developments, earlier experimental work, the theory or the law governing the phenomenon being investigated. These introductory remarks must be kept brief to avoid obscuring the main point of the investigation. Sometimes, you will be asked to carry out an investigation in order to dispute or challenge a claim that has been made against a generally accepted scientific phenomenon. If that is the case, state the currently held theory and the claim that has been made. Your introduction must state the aim of your investigation. The integration of the aim into the introduction allows for the smooth transition from general information to the specific goal of the investigation.

Apparatus and method:

Present a clear, concise and step-by-step description of the apparatus, techniques and procedures used. List and name the apparatus with brief descriptions of the main parts as well as the functions thereof. A neatly sketched and labelled **diagram** of the apparatus is essential and can save you paragraphs of tedious written descriptions. Briefly describe the procedures that you followed in the investigation. Wherever appropriate, give a reason for each step you took in a procedure. Sometimes more than one section is required for this material. For example, if two techniques were used, i.e. Technique No.1, Technique No.2, etc. then a brief explanation of each of these techniques must be given. Do not omit any significant steps. A description of the method helps you to recall the problems that were associated with experimental procedures such as precision of instruments, strengths and weaknesses of certain techniques, recording of observations etc. This will help you when you have to summarise your conclusions and recommendations at the end of your report.

Data and results:

The main part of this section are your tables of results and graphs. You must have a consistent way of recording your observations and calculations. Data are normally summarised and displayed in tables and graphs. Each table and graph is usually referenced by a number and should be numbered in sequence, e.g. Table 1, Table 2, Figure 1, Figure 2 etc. Each table is accompanied by a title and each graph by a caption which describes the purpose for which it has been presented. (e.g. “Table 1: Measurements of the width of the cylinder” and “Graph 3: To determine the viscosity of the sample of oil”). Tables and figures must be referred to in the text, e.g. “The apparatus was arranged as shown in Figure 1”; “The data gathered were recorded as shown in Table 1 below” or “The data in Table 1 were used to plot the graphs in Figure 1, 2 and 3.” These brief statements help to link the different parts of the report. This section does not contain your judgement of the data. It is a straightforward presentation of your readings or measurements.

Analysis, interpretation and discussion:

In this section examine and extract important aspects of the data and use these to explain various relationships or determine your measurement results. For example, what is the shape of your graph and what does it suggest in terms of the relationship between the variables? If it is a straight line, what is the value of the slope? Remember to quote your results with the appropriate significant figures and the corresponding uncertainties. How does an expected value compare with your own and what reasons can you give for this? You may give tentative explanations for your data but be careful not to mix facts with opinions.

Conclusion and recommendations:

This is normally a section in which you say what the investigation has shown and to what extent the problem or claim stated in the introduction has been resolved. Remember that any conclusions must be supported by evidence from your data. Always quote any final results, together with their uncertainties in your conclusion. Avoid making vague statements such as “This was a successful experiment.” You may also need to discuss sources of uncertainty and any improvements that could be made to the apparatus (and measurements). Again, avoid meaningless phrases such as “it was caused by human error.”

Writing a report allows you to reflect critically on the whole experiment and check your understanding of the purpose of the investigation as well as produce an accurate record of it. Note that a physics practical is not a set of procedures designed to reproduce some “correct” answer. It is a problem that has been posed that requires an experimental solution which may include making measurements, implementing different procedures and techniques and then the formulation of a suitable report.

F.2 Scientific style

Very often, in reports of this kind, writers prefer to use the passive construction or impersonal style to report procedures followed in conducting experiments, by writing, for example: “Five measurements were taken,” instead of “I / We took five measurements.”

Both styles are acceptable. As you do most of your practicals in groups, you are likely to visualise what you did as a group and report it as a group activity in which case the personal pronoun “we” is appropriate. However, once you have chosen a style of writing, then you must use it throughout your report and not switch back and forth between the two.

F.4 A laboratory report checklist

When preparing your report, use the following checklist to see whether or not you have included everything correctly.

Introduction and aim:

- ☒ Have I written my name, my partners' names and my course at the top of my report?
- ☒ Does my report have a title and a date?
- ☒ Does my aim clearly outline the purpose(s) of my experiment?
- ☒ Have I clearly outlined the theory involved in my investigation?

Apparatus:

- ☒ Have I listed all the apparatus (with a diagram)?
- ☒ Is my diagram labelled and does it have a heading?

Method:

- ☒ Have I described how I went about my experiment (i.e. what I did)?
- ☒ Have I clearly described the significant steps in the procedure (including the analysis on my data)?
- ☒ Have I explained why certain aspects of the procedure were undertaken?

Data / Results:

- ☒ Have I recorded all relevant measurements?
- ☒ Do all my tables have titles?
- ☒ Do all the columns in my tables have headings with units?
- ☒ Are all my data recorded correctly?

Graphs:

- ☒ Does my graph have a caption which should state why I plotted my graph?
- ☒ Have I used a decent scale so my graph fills the whole page?
- ☒ Have I labelled the axes, including the correct units?
- ☒ Have I used a $\odot$ for the data points and not a $\bullet$ (blob)?
- ☒ Have I used pencil for my graph (and not pen)?
- ☒ Have I drawn the best straight line through my data points? (if necessary)
- ☒ Have I shown how I got the slope from my line? (if necessary)
- ☒ Have I recorded the slope and intercept (with uncertainties) correctly? (if necessary)

Analysis and discussion:

- ☒ Have I described and explained important relationships revealed by the data (and graphs)?
- ☒ Have I clearly set out my calculations?
- ☒ Are my calculations in the correct section of the report?
- ☒ Have I used SI units throughout?
- ☒ Have I shown how I calculated the uncertainties in the results?
- ☒ Is my final result presented correctly with the appropriate significant figures?
- ☒ Have I described and explained the results of my calculations?

Conclusion and recommendations:

- ☒ Does my conclusion refer back to my aims for doing the experiment?
- ☒ Have I fulfilled all my aims for doing the experiment?
- ☒ Have I quoted all my results with their uncertainties?
- ☒ Have I recommended how this experiment may be improved?